

作业 2 编程部分

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编程部分

1 卷积神经网络分类任务代码补充

1.1 datasets.py: 数据预处理变换、数据集、数据加载器定义

为测试数据加载与定义的数据增广变换能否正常运行，执行 `python unit_test.py data_loader`:

```
(demo) D:\college\media\HW2-B\code>python unit_test.py data_loader  
X N F U M S B T
```

图 1: 加载测试数据

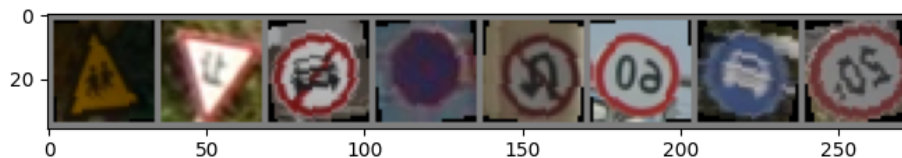


图 2: 存储加载的测试数据图片

由输出可知，8 个样本被顺利载入，并完成了数据增广变换。

1.2 networks.py: 卷积块和分类器模型的定义

为测试 STN 网络初始化是否正确，执行 `python unit_test.py stn`:

```
(demo) D:\college\media\HW2-B\code>python unit_test.py stn  
STN forward check passed.
```

图 3: 验证 STN 网络初始化

通过验证，说明 STN 网络初始化正确。

2 训练/测试/可视化

2.1 探究 batch normalization 和 dropout 的作用

2.1.1 默认配置模型

使用默认配置训练模型：

执行 `python train.py --ckpt_path checkpoints/default`

```
(demo) D:\college\media\HW2-B\code>python train.py --ckpt_path checkpoints/default
training...
Epoch 1/15: 100% | 65/65 [00:09<00:00, 6.85it/s]
Epoch 01: loss = 2.836, accuracy on validation set = 0.284
Model saved in checkpoints/default\ckpt_epoch_1.pth

Epoch 2/15: 100% | 65/65 [00:03<00:00, 18.53it/s]
Epoch 02: loss = 2.256, accuracy on validation set = 0.279
Model saved in checkpoints/default\ckpt_epoch_2.pth

Epoch 3/15: 100% | 65/65 [00:03<00:00, 16.41it/s]
Epoch 03: loss = 1.891, accuracy on validation set = 0.408
Model saved in checkpoints/default\ckpt_epoch_3.pth

Epoch 4/15: 100% | 65/65 [00:04<00:00, 15.99it/s]
Epoch 04: loss = 1.466, accuracy on validation set = 0.533
Model saved in checkpoints/default\ckpt_epoch_4.pth

Epoch 5/15: 100% | 65/65 [00:03<00:00, 16.70it/s]
Epoch 05: loss = 1.187, accuracy on validation set = 0.554
Model saved in checkpoints/default\ckpt_epoch_5.pth

Epoch 6/15: 100% | 65/65 [00:03<00:00, 16.31it/s]
Epoch 06: loss = 0.942, accuracy on validation set = 0.627
Model saved in checkpoints/default\ckpt_epoch_6.pth

Epoch 7/15: 100% | 65/65 [00:04<00:00, 16.03it/s]
Epoch 07: loss = 0.799, accuracy on validation set = 0.665
Model saved in checkpoints/default\ckpt_epoch_7.pth

Epoch 8/15: 100% | 65/65 [00:03<00:00, 18.25it/s]
Epoch 08: loss = 0.627, accuracy on validation set = 0.744
Model saved in checkpoints/default\ckpt_epoch_8.pth

Epoch 9/15: 100% | 65/65 [00:03<00:00, 19.90it/s]
Epoch 09: loss = 0.474, accuracy on validation set = 0.738
Model saved in checkpoints/default\ckpt_epoch_9.pth

Epoch 10/15: 100% | 65/65 [00:03<00:00, 18.46it/s]
Epoch 10: loss = 0.394, accuracy on validation set = 0.742
Model saved in checkpoints/default\ckpt_epoch_10.pth

Epoch 11/15: 100% | 65/65 [00:04<00:00, 14.51it/s]
Epoch 11: loss = 0.384, accuracy on validation set = 0.763
Model saved in checkpoints/default\ckpt_epoch_11.pth

Epoch 12/15: 100% | 65/65 [00:05<00:00, 11.58it/s]
Epoch 12: loss = 0.249, accuracy on validation set = 0.773
Model saved in checkpoints/default\ckpt_epoch_12.pth

Epoch 13/15: 100% | 65/65 [00:09<00:00, 6.99it/s]
Epoch 13: loss = 0.187, accuracy on validation set = 0.773
Model saved in checkpoints/default\ckpt_epoch_13.pth

Epoch 14/15: 100% | 65/65 [00:05<00:00, 11.82it/s]
Epoch 14: loss = 0.190, accuracy on validation set = 0.798
Model saved in checkpoints/default\ckpt_epoch_14.pth

Epoch 15/15: 100% | 65/65 [00:05<00:00, 12.20it/s]
Epoch 15: loss = 0.144, accuracy on validation set = 0.752
Model saved in checkpoints/default\ckpt_epoch_15.pth
```

图 4: 使用默认配置的训练过程

使用默认配置训练模型时，验证集的准确率最高为 79.8%。

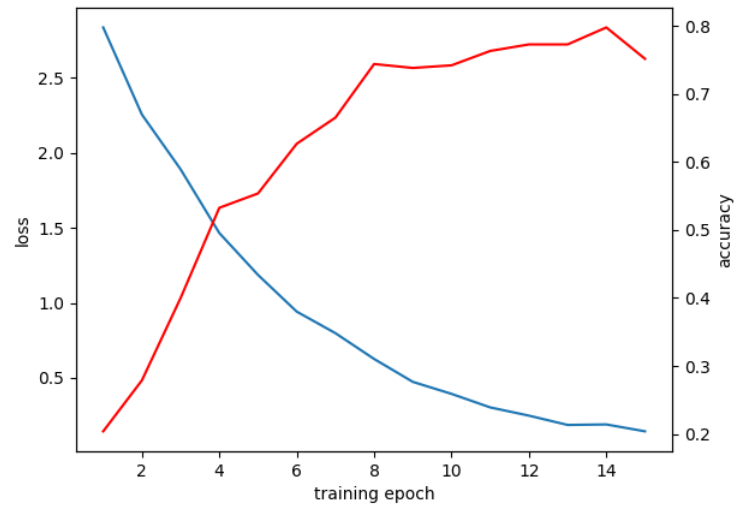


图 5: 使用默认配置时的损失函数

最优模型为验证集错误率最低点，由损失函数曲线可知使用默认配置时轮数 epoch=14 为最优模型。

测试默认配置模型：

执行 `python test.py -ckpt_path checkpoints/default -epoch 14`

```
(demo) D:\college\media\HW2-8\code>python test.py --ckpt_path checkpoints/default --epoch 14
[Info] loading checkpoint from checkpoints/default\ckpt_epoch_14.pth ...
accuracy on the test set: 0.788
```

图 6: 默认配置模型的测试集准确率

使用默认配置训练模型时，测试集的准确率为 78.8%，比验证集的最高准确率低 1%。

2.1.2 启用 batch normalization 的模型

启用 batch normalization 训练模型：

执行 `python train.py -ckpt_path checkpoints/bn -bn`

```
(demo) D:\college\media\HW2-B\code>python train.py --ckpt_path checkpoints/bn --bn
training...
Epoch 1/15: 100% | 65/65 [00:18<00:00, 3.56it/s]
Epoch 01: loss = 1.916, accuracy on validation set = 0.667
Model saved in checkpoints/bn\ckpt_epoch_1.pth

Epoch 2/15: 100% | 65/65 [00:04<00:00, 13.69it/s]
Epoch 02: loss = 0.785, accuracy on validation set = 0.752
Model saved in checkpoints/bn\ckpt_epoch_2.pth

Epoch 3/15: 100% | 65/65 [00:05<00:00, 12.42it/s]
Epoch 03: loss = 0.436, accuracy on validation set = 0.752
Model saved in checkpoints/bn\ckpt_epoch_3.pth

Epoch 4/15: 100% | 65/65 [00:05<00:00, 12.97it/s]
Epoch 04: loss = 0.247, accuracy on validation set = 0.831
Model saved in checkpoints/bn\ckpt_epoch_4.pth

Epoch 5/15: 100% | 65/65 [00:04<00:00, 13.71it/s]
Epoch 05: loss = 0.161, accuracy on validation set = 0.821
Model saved in checkpoints/bn\ckpt_epoch_5.pth

Epoch 6/15: 100% | 65/65 [00:04<00:00, 15.25it/s]
Epoch 06: loss = 0.140, accuracy on validation set = 0.802
Model saved in checkpoints/bn\ckpt_epoch_6.pth

Epoch 7/15: 100% | 65/65 [00:03<00:00, 17.28it/s]
Epoch 07: loss = 0.083, accuracy on validation set = 0.867
Model saved in checkpoints/bn\ckpt_epoch_7.pth

Epoch 8/15: 100% | 65/65 [00:03<00:00, 16.56it/s]
Epoch 08: loss = 0.032, accuracy on validation set = 0.890
Model saved in checkpoints/bn\ckpt_epoch_8.pth

Epoch 9/15: 100% | 65/65 [00:04<00:00, 14.60it/s]
Epoch 09: loss = 0.016, accuracy on validation set = 0.908
Model saved in checkpoints/bn\ckpt_epoch_9.pth

Epoch 10/15: 100% | 65/65 [00:05<00:00, 12.30it/s]
Epoch 10: loss = 0.010, accuracy on validation set = 0.906
Model saved in checkpoints/bn\ckpt_epoch_10.pth

Epoch 11/15: 100% | 65/65 [00:05<00:00, 12.00it/s]
Epoch 11: loss = 0.012, accuracy on validation set = 0.910
Model saved in checkpoints/bn\ckpt_epoch_11.pth

Epoch 12/15: 100% | 65/65 [00:05<00:00, 12.18it/s]
Epoch 12: loss = 0.013, accuracy on validation set = 0.887
Model saved in checkpoints/bn\ckpt_epoch_12.pth

Epoch 13/15: 100% | 65/65 [00:04<00:00, 13.32it/s]
Epoch 13: loss = 0.012, accuracy on validation set = 0.906
Model saved in checkpoints/bn\ckpt_epoch_13.pth

Epoch 14/15: 100% | 65/65 [00:05<00:00, 11.98it/s]
Epoch 14: loss = 0.007, accuracy on validation set = 0.919
Model saved in checkpoints/bn\ckpt_epoch_14.pth

Epoch 15/15: 100% | 65/65 [00:05<00:00, 11.80it/s]
Epoch 15: loss = 0.003, accuracy on validation set = 0.923
Model saved in checkpoints/bn\ckpt_epoch_15.pth
```

图 7: 启用 batch normalization 的训练过程

启用 batch normalization 训练模型时，验证集的准确率最高为 92.3%，较默认配置时准确率有明显提升。但损失函数曲线的波动较大，存在过拟合现象。batch normalization 作为一种归一化方法，可以避免梯度消失和梯度溢出，并加速收敛，可提升泛化能力。

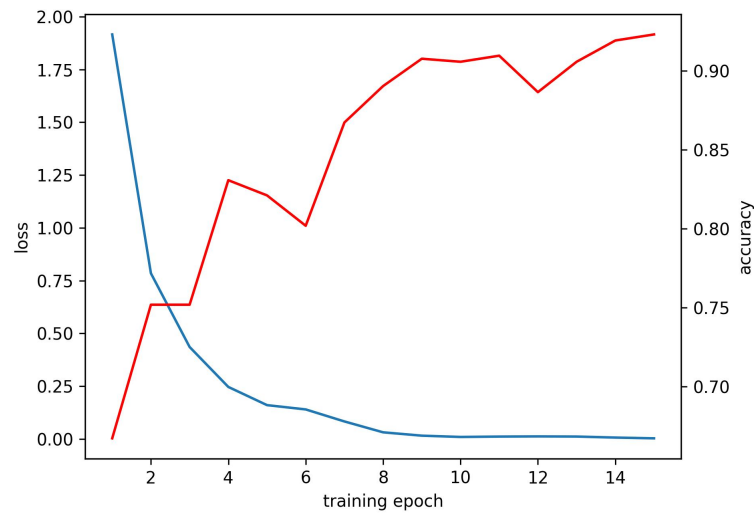


图 8: 启用 batch normalization 的损失函数

最优模型为验证集错误率最低点，由损失函数曲线可知启用 batch normalization 时轮数 epoch=15 为最优模型。

测试启用 batch normalization 的模型：

执行 `python test.py -ckpt_path checkpoints/bn -epoch 15`

```
(demo) D:\college\media\HW2-B\code>python test.py --ckpt_path checkpoints/bn --epoch 15
[Info] loading checkpoint from checkpoints/bn\ckpt_epoch_15.pth ...
accuracy on the test set: 0.901
```

图 9: 启用 batch normalization 的测试集准确率

启用 batch normalization 训练模型时，测试集的准确率为 90.1%，比验证集的最高准确率低 2.2%，但仍可以看出引入 batch normalization 的模型性能相较默认配置有明显提升。

2.1.3 启用 dropout（概率为 0.4）的模型

启用 dropout（概率为 0.4）训练模型：

执行 `python train.py -ckpt_path checkpoints/dropout --dropout`

```
(demo) D:\college\media\HW2-8\code>python train.py --ckpt_path checkpoints/dropout --dropout 0.4
training...
Epoch 1/15: 100% | 65/65 [00:05<00:00, 12.00it/s]
Epoch 01: loss = 3.010, accuracy on validation set = 0.135
Model saved in checkpoints/dropout\ckpt_epoch_1.pth

Epoch 2/15: 100% | 65/65 [00:04<00:00, 15.53it/s]
Epoch 02: loss = 2.547, accuracy on validation set = 0.219
Model saved in checkpoints/dropout\ckpt_epoch_2.pth

Epoch 3/15: 100% | 65/65 [00:03<00:00, 17.49it/s]
Epoch 03: loss = 2.251, accuracy on validation set = 0.281
Model saved in checkpoints/dropout\ckpt_epoch_3.pth

Epoch 4/15: 100% | 65/65 [00:04<00:00, 13.51it/s]
Epoch 04: loss = 1.999, accuracy on validation set = 0.387
Model saved in checkpoints/dropout\ckpt_epoch_4.pth

Epoch 5/15: 100% | 65/65 [00:14<00:00, 4.64it/s]
Epoch 05: loss = 1.770, accuracy on validation set = 0.425
Model saved in checkpoints/dropout\ckpt_epoch_5.pth

Epoch 6/15: 100% | 65/65 [00:09<00:00, 6.55it/s]
Epoch 06: loss = 1.582, accuracy on validation set = 0.410
Model saved in checkpoints/dropout\ckpt_epoch_6.pth

Epoch 7/15: 100% | 65/65 [00:03<00:00, 18.00it/s]
Epoch 07: loss = 1.470, accuracy on validation set = 0.479
Model saved in checkpoints/dropout\ckpt_epoch_7.pth

Epoch 8/15: 100% | 65/65 [00:10<00:00, 6.29it/s]
Epoch 08: loss = 1.228, accuracy on validation set = 0.596
Model saved in checkpoints/dropout\ckpt_epoch_8.pth

Epoch 9/15: 100% | 65/65 [00:13<00:00, 4.88it/s]
Epoch 09: loss = 1.156, accuracy on validation set = 0.602
Model saved in checkpoints/dropout\ckpt_epoch_9.pth

Epoch 10/15: 100% | 65/65 [00:12<00:00, 5.22it/s]
Epoch 10: loss = 0.972, accuracy on validation set = 0.652
Model saved in checkpoints/dropout\ckpt_epoch_10.pth

Epoch 11/15: 100% | 65/65 [00:14<00:00, 4.60it/s]
Epoch 11: loss = 0.911, accuracy on validation set = 0.683
Model saved in checkpoints/dropout\ckpt_epoch_11.pth

Epoch 12/15: 100% | 65/65 [00:05<00:00, 11.20it/s]
Epoch 12: loss = 0.869, accuracy on validation set = 0.715
Model saved in checkpoints/dropout\ckpt_epoch_12.pth

Epoch 13/15: 100% | 65/65 [00:06<00:00, 10.66it/s]
Epoch 13: loss = 0.731, accuracy on validation set = 0.708
Model saved in checkpoints/dropout\ckpt_epoch_13.pth

Epoch 14/15: 100% | 65/65 [00:05<00:00, 11.01it/s]
Epoch 14: loss = 0.655, accuracy on validation set = 0.744
Model saved in checkpoints/dropout\ckpt_epoch_14.pth

Epoch 15/15: 100% | 65/65 [00:16<00:00, 3.86it/s]
Epoch 15: loss = 0.505, accuracy on validation set = 0.733
Model saved in checkpoints/dropout\ckpt_epoch_15.pth
```

图 10: 启用 dropout（概率为 0.4）的训练过程

启用 dropout（概率为 0.4）训练模型时，验证集的准确率最高为 74.4%，较默认配置时准确率略有下降，但损失函数曲线的波动较小。这是因为 dropout 作为一种正则化方法，在训练过程中每次迭代时，会将各层节点的激活值以一定概率随机置 0，可有效防止过拟合，提升模型的泛化能力。

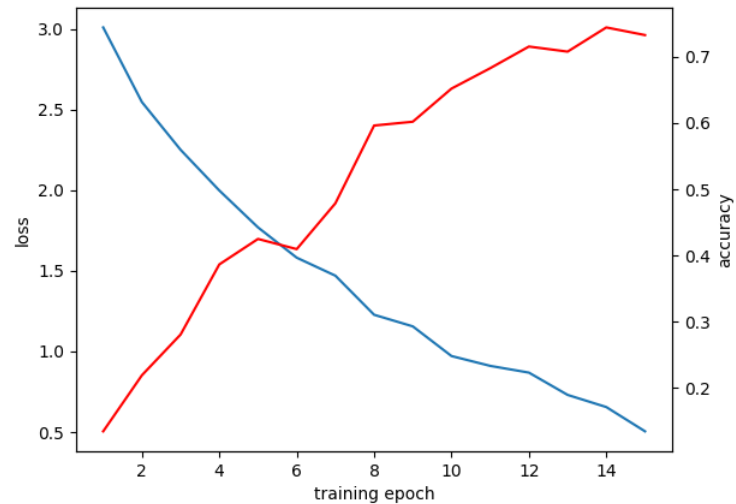


图 11: 启用 dropout (概率为 0.4) 的损失函数

最优模型为验证集错误率最低点, 由损失函数曲线可知启用 dropout (概率为 0.4) 时轮数 epoch=14 为最优模型。

测试启用 dropout (概率为 0.4) 的模型:

执行 `python test.py -ckpt_path checkpoints/dropout -epoch 14`

```
(demo) D:\college\media\HW2-B\code>python test.py --ckpt_path checkpoints/dropout --epoch 14
[Info] loading checkpoint from checkpoints/dropout\ckpt_epoch_14.pth ...
accuracy on the test set: 0.732
```

图 12: 启用 dropout (概率为 0.4) 的测试集准确率

启用 dropout (概率为 0.4) 训练模型时, 测试集的准确率为 73.2%, 比验证集的最高准确率低 1.2%。

2.2 探究模型深度的影响

2.2.1 ConvBlock 的数量为 6

训练 6 个 ConvBlock 的模型：

执行 `python train.py -ckpt_path checkpoints/bn_6convs -bn`

```
(demo) D:\college\media\HW2-8\code>python train.py --ckpt_path checkpoints/bn_6convs --bn
training...
Epoch 1/15: 100% | 65/65 [00:05<00:00, 11.17it/s]
Epoch 01: loss = 1.962, accuracy on validation set = 0.648
Model saved in checkpoints/bn_6convs/ckpt_epoch_1.pth

Epoch 2/15: 100% | 65/65 [00:05<00:00, 12.19it/s]
Epoch 02: loss = 0.872, accuracy on validation set = 0.658
Model saved in checkpoints/bn_6convs/ckpt_epoch_2.pth

Epoch 3/15: 100% | 65/65 [00:05<00:00, 12.08it/s]
Epoch 03: loss = 0.497, accuracy on validation set = 0.781
Model saved in checkpoints/bn_6convs/ckpt_epoch_3.pth

Epoch 4/15: 100% | 65/65 [00:06<00:00, 10.37it/s]
Epoch 04: loss = 0.315, accuracy on validation set = 0.840
Model saved in checkpoints/bn_6convs/ckpt_epoch_4.pth

Epoch 5/15: 100% | 65/65 [00:05<00:00, 11.13it/s]
Epoch 05: loss = 0.204, accuracy on validation set = 0.810
Model saved in checkpoints/bn_6convs/ckpt_epoch_5.pth

Epoch 6/15: 100% | 65/65 [00:04<00:00, 13.75it/s]
Epoch 06: loss = 0.112, accuracy on validation set = 0.867
Model saved in checkpoints/bn_6convs/ckpt_epoch_6.pth

Epoch 7/15: 100% | 65/65 [00:04<00:00, 14.77it/s]
Epoch 07: loss = 0.080, accuracy on validation set = 0.852
Model saved in checkpoints/bn_6convs/ckpt_epoch_7.pth

Epoch 8/15: 100% | 65/65 [00:04<00:00, 14.26it/s]
Epoch 08: loss = 0.071, accuracy on validation set = 0.852
Model saved in checkpoints/bn_6convs/ckpt_epoch_8.pth

Epoch 9/15: 100% | 65/65 [00:05<00:00, 11.91it/s]
Epoch 09: loss = 0.037, accuracy on validation set = 0.863
Model saved in checkpoints/bn_6convs/ckpt_epoch_9.pth

Epoch 10/15: 100% | 65/65 [00:05<00:00, 11.33it/s]
Epoch 10: loss = 0.030, accuracy on validation set = 0.892
Model saved in checkpoints/bn_6convs/ckpt_epoch_10.pth

Epoch 11/15: 100% | 65/65 [00:05<00:00, 10.98it/s]
Epoch 11: loss = 0.019, accuracy on validation set = 0.892
Model saved in checkpoints/bn_6convs/ckpt_epoch_11.pth

Epoch 12/15: 100% | 65/65 [00:05<00:00, 11.78it/s]
Epoch 12: loss = 0.006, accuracy on validation set = 0.902
Model saved in checkpoints/bn_6convs/ckpt_epoch_12.pth

Epoch 13/15: 100% | 65/65 [00:05<00:00, 11.27it/s]
Epoch 13: loss = 0.003, accuracy on validation set = 0.910
Model saved in checkpoints/bn_6convs/ckpt_epoch_13.pth

Epoch 14/15: 100% | 65/65 [00:06<00:00, 10.31it/s]
Epoch 14: loss = 0.002, accuracy on validation set = 0.925
Model saved in checkpoints/bn_6convs/ckpt_epoch_14.pth

Epoch 15/15: 100% | 65/65 [00:06<00:00, 10.39it/s]
Epoch 15: loss = 0.002, accuracy on validation set = 0.921
Model saved in checkpoints/bn_6convs/ckpt_epoch_15.pth
```

图 13: 6 个 ConvBlock 的模型的训练过程

ConvBlock 数量为 6 的模型，其验证集的准确率最高为 92.5%，准确率较 ConvBlock 数量为 5 的初始情况有所提升，损失函数曲线的波动较大，也存在过拟合现象。

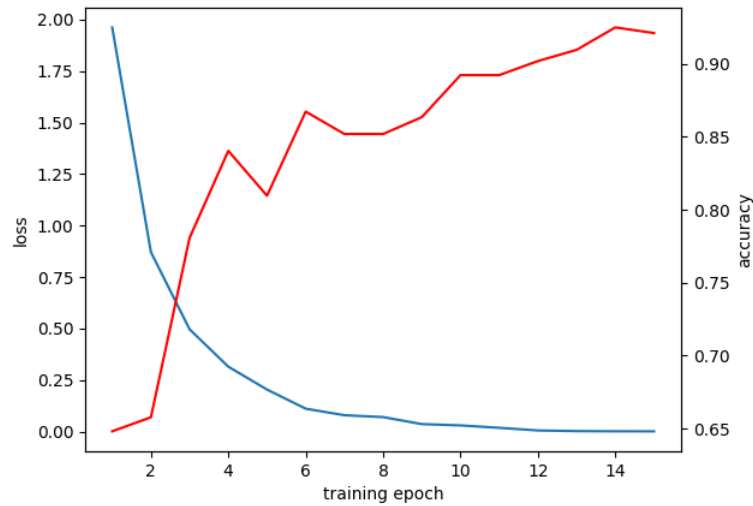


图 14: 6 个 ConvBlock 的模型的损失函数

测试 6 个 ConvBlock 的模型:

执行 `python test.py -ckpt_path checkpoints/bn_6convs`

```
(demo) D:\college\media\HW2-B\code>python test.py --ckpt_path checkpoints/bn_6convs --epoch 15
[Info] loading checkpoint from checkpoints/bn_6convs\ckpt_epoch_15.pth ...
accuracy on the test set: 0.914
```

图 15: 6 个 ConvBlock 的模型的测试集准确率

ConvBlock 数量为 6 的模型，其测试集的准确率为 91.4%，比验证集的最高准确率低 1.1%，较 ConvBlock 数量为 5 的初始情况略有提升。

2.2.2 ConvBlock 的数量为 7

训练 7 个 ConvBlock 的模型：

执行 `python train.py -ckpt_path checkpoints/bn_7convs -bn`

```
(demo) D:\college\media\HW2-B\code>python train.py --ckpt_path checkpoints/bn_7convs -bn
training...
Epoch 1/15: 100% | 65/65 [00:06<00:00, 10.03it/s]
Epoch 01: loss = 2.185, accuracy on validation set = 0.446
Model saved in checkpoints/bn_7convs\ckpt_epoch_1.pth

Epoch 2/15: 100% | 65/65 [00:11<00:00, 5.54it/s]
Epoch 02: loss = 1.051, accuracy on validation set = 0.737
Model saved in checkpoints/bn_7convs\ckpt_epoch_2.pth

Epoch 3/15: 100% | 65/65 [00:20<00:00, 3.19it/s]
Epoch 03: loss = 0.635, accuracy on validation set = 0.767
Model saved in checkpoints/bn_7convs\ckpt_epoch_3.pth

Epoch 4/15: 100% | 65/65 [00:24<00:00, 2.67it/s]
Epoch 04: loss = 0.481, accuracy on validation set = 0.746
Model saved in checkpoints/bn_7convs\ckpt_epoch_4.pth

Epoch 5/15: 100% | 65/65 [00:23<00:00, 2.73it/s]
Epoch 05: loss = 0.241, accuracy on validation set = 0.854
Model saved in checkpoints/bn_7convs\ckpt_epoch_5.pth

Epoch 6/15: 100% | 65/65 [00:21<00:00, 3.00it/s]
Epoch 06: loss = 0.175, accuracy on validation set = 0.815
Model saved in checkpoints/bn_7convs\ckpt_epoch_6.pth

Epoch 7/15: 100% | 65/65 [00:18<00:00, 3.49it/s]
Epoch 07: loss = 0.140, accuracy on validation set = 0.837
Model saved in checkpoints/bn_7convs\ckpt_epoch_7.pth

Epoch 8/15: 100% | 65/65 [00:23<00:00, 2.73it/s]
Epoch 08: loss = 0.102, accuracy on validation set = 0.852
Epoch 9/15: 100% | 65/65 [00:21<00:00, 3.06it/s]
Epoch 09: loss = 0.046, accuracy on validation set = 0.877
Model saved in checkpoints/bn_7convs\ckpt_epoch_9.pth

Epoch 10/15: 100% | 65/65 [00:06<00:00, 9.60it/s]
Epoch 10: loss = 0.030, accuracy on validation set = 0.875
Model saved in checkpoints/bn_7convs\ckpt_epoch_10.pth

Epoch 11/15: 100% | 65/65 [00:06<00:00, 9.50it/s]
Epoch 11: loss = 0.036, accuracy on validation set = 0.877
Model saved in checkpoints/bn_7convs\ckpt_epoch_11.pth

Epoch 12/15: 100% | 65/65 [00:07<00:00, 9.26it/s]
Epoch 12: loss = 0.013, accuracy on validation set = 0.898
Model saved in checkpoints/bn_7convs\ckpt_epoch_12.pth

Epoch 13/15: 100% | 65/65 [00:06<00:00, 9.59it/s]
Epoch 13: loss = 0.006, accuracy on validation set = 0.921
Model saved in checkpoints/bn_7convs\ckpt_epoch_13.pth

Epoch 14/15: 100% | 65/65 [00:07<00:00, 8.96it/s]
Epoch 14: loss = 0.003, accuracy on validation set = 0.919
Model saved in checkpoints/bn_7convs\ckpt_epoch_14.pth

Epoch 15/15: 100% | 65/65 [00:07<00:00, 8.19it/s]
Epoch 15: loss = 0.002, accuracy on validation set = 0.917
Model saved in checkpoints/bn_7convs\ckpt_epoch_15.pth
```

图 16: 7 个 ConvBlock 的模型的训练过程

ConvBlock 数量为 7 的模型，其验证集的准确率最高为 92.1%，准确率较 ConvBlock 数量为 5 的初始情况有所提升，较 ConvBlock 数量为 6 的情况略有下降，损失函数曲线的波动较大，也存在过拟合现象。

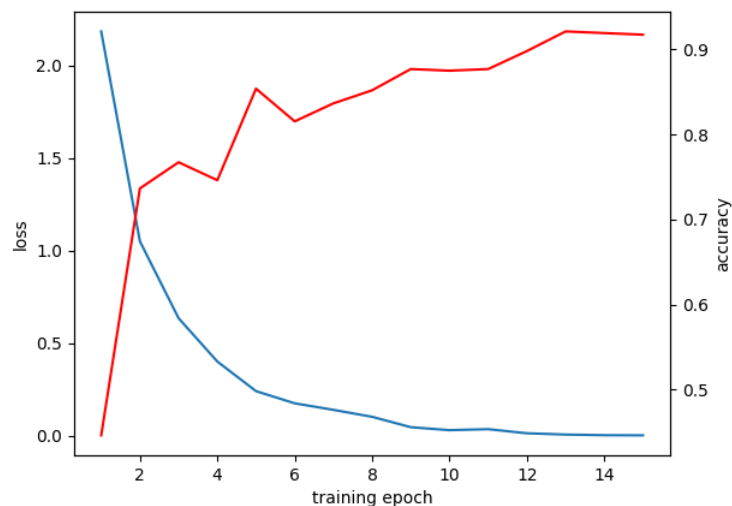


图 17: 7 个 ConvBlock 的模型的损失函数

测试 7 个 ConvBlock 的模型:

执行 `python test.py -ckpt_path checkpoints/bn_7convs`

```
(demo) D:\college\media\HW2-B\code>python test.py --ckpt_path checkpoints/bn_7convs --epoch 15  
[Info] loading checkpoint from checkpoints/bn_7convs\ckpt_epoch_15.pth ...  
accuracy on the test set: 0.912
```

图 18: 7 个 ConvBlock 的模型的测试集准确率

ConvBlock 数量为 7 的模型，其测试集的准确率为 91.2%，比验证集的最高准确率低 0.9%，较 ConvBlock 数量为 5 的初始情况略有提升，较 ConvBlock 数量为 6 的情况略有下降，但验证集最高准确率与测试集准确率之差最小，说明泛化能力增强。

综合 ConvBlock 数量为 5、6、7 的三种情况可知，在一定程度上增加模型深度可以提升测试集的准确率与验证集的最高准确率，提升模型性能；且模型深度的增加还会提升模型的泛化能力。

但在数据集规模较小时，模型均存在过拟合现象，深层模型的过拟合现象可能会更严重，因此应选择合适的模型深度以平衡模型的性能、泛化能力以及过拟合问题。

2.3 探究数据增广的作用

已在报告 1.1 部分附上了 datasets.py 中数据增广变换的输出结果。

在代码中使用了如下 3 种增广变换，涉及了图像的旋转、翻转、颜色变化，覆盖的变换种类较广：

随机小角度（小于 15° ）旋转

```
data_transform.insert(1, transforms.RandomRotation(15))
```

随机水平翻转

```
data_transform.insert(1, transforms.RandomHorizontalFlip())
```

颜色扰动（亮度、对比度、饱和度的微调）

```
data_transform.insert(1, transforms.ColorJitter(brightness=0.2,  
contrast=0.2, saturation=0.2))
```

训练加入数据增广的预处理变换的模型：

执行 `python train.py -ckpt_path checkpoints/bn_aug -bn -augment`

```
(demo) D:\college\media\HW2-8\code>python train.py --ckpt_path checkpoints/bn_aug --bn --augment  
training...  
Epoch 1/15: 100% | 65/65 [00:00<00:00, 7.46it/s]  
Epoch 01: loss = 2.259, accuracy on validation set = 0.498  
Model saved in checkpoints/bn_aug\ckpt_epoch_1.pth  
  
Epoch 2/15: 100% | 65/65 [00:04<00:00, 16.17it/s]  
Epoch 02: loss = 1.368, accuracy on validation set = 0.640  
Model saved in checkpoints/bn_aug\ckpt_epoch_2.pth  
  
Epoch 3/15: 100% | 65/65 [00:04<00:00, 15.68it/s]  
Epoch 03: loss = 0.970, accuracy on validation set = 0.665  
Model saved in checkpoints/bn_aug\ckpt_epoch_3.pth  
  
Epoch 4/15: 100% | 65/65 [00:04<00:00, 15.63it/s]  
Epoch 04: loss = 0.789, accuracy on validation set = 0.719  
Model saved in checkpoints/bn_aug\ckpt_epoch_4.pth  
  
Epoch 5/15: 100% | 65/65 [00:04<00:00, 15.59it/s]  
Epoch 05: loss = 0.643, accuracy on validation set = 0.738  
Model saved in checkpoints/bn_aug\ckpt_epoch_5.pth  
  
Epoch 6/15: 100% | 65/65 [00:04<00:00, 15.90it/s]  
Epoch 06: loss = 0.582, accuracy on validation set = 0.752  
Model saved in checkpoints/bn_aug\ckpt_epoch_6.pth  
  
Epoch 7/15: 100% | 65/65 [00:03<00:00, 17.31it/s]  
Epoch 07: loss = 0.493, accuracy on validation set = 0.771  
Model saved in checkpoints/bn_aug\ckpt_epoch_7.pth  
  
Epoch 8/15: 100% | 65/65 [00:03<00:00, 17.35it/s]  
Epoch 08: loss = 0.397, accuracy on validation set = 0.821  
Model saved in checkpoints/bn_aug\ckpt_epoch_8.pth  
  
Epoch 9/15: 100% | 65/65 [00:03<00:00, 16.61it/s]  
Epoch 09: loss = 0.409, accuracy on validation set = 0.798  
Model saved in checkpoints/bn_aug\ckpt_epoch_9.pth  
  
Epoch 10/15: 100% | 65/65 [00:03<00:00, 17.04it/s]  
Epoch 10: loss = 0.348, accuracy on validation set = 0.821  
Model saved in checkpoints/bn_aug\ckpt_epoch_10.pth  
  
Epoch 11/15: 100% | 65/65 [00:04<00:00, 15.22it/s]  
Epoch 11: loss = 0.280, accuracy on validation set = 0.860  
Model saved in checkpoints/bn_aug\ckpt_epoch_11.pth  
  
Epoch 12/15: 100% | 65/65 [00:04<00:00, 14.44it/s]  
Epoch 12: loss = 0.233, accuracy on validation set = 0.867  
Model saved in checkpoints/bn_aug\ckpt_epoch_12.pth  
  
Epoch 13/15: 100% | 65/65 [00:04<00:00, 13.96it/s]  
Epoch 13: loss = 0.220, accuracy on validation set = 0.862  
Model saved in checkpoints/bn_aug\ckpt_epoch_13.pth  
  
Epoch 14/15: 100% | 65/65 [00:04<00:00, 14.28it/s]  
Epoch 14: loss = 0.196, accuracy on validation set = 0.888  
Model saved in checkpoints/bn_aug\ckpt_epoch_14.pth  
  
Epoch 15/15: 100% | 65/65 [00:04<00:00, 14.96it/s]  
Epoch 15: loss = 0.200, accuracy on validation set = 0.869  
Model saved in checkpoints/bn_aug\ckpt_epoch_15.pth
```

图 19: 加入数据增广的预处理变换后模型的训练过程

加入数据增广预处理变换训练模型时，验证集的准确率最高为 88.8%，较默认配置时准确率有所提升，且损失函数曲线的波动变小。通过简单的几何和颜色变换，数据增广预处理增加了数据多样性，有效缓解过拟合，提升模型的泛化能力。

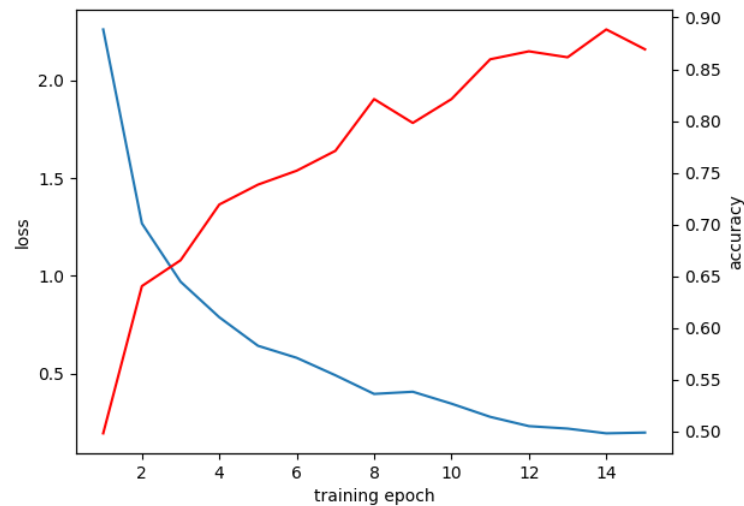


图 20: 加入数据增广的预处理变换后模型的损失函数

最优模型为验证集错误率最低点，由损失函数曲线可知加入数据增广的预处理变换后轮数 epoch=14 为最优模型。

测试加入数据增广的预处理变换的模型：

执行 `python test.py -ckpt_path checkpoints/bn_aug -epoch`

```
(demo) D:\college\media\HW2-B\code>python test.py --ckpt_path checkpoints/bn_aug --epoch 14
[Info] loading checkpoint from checkpoints/bn_aug\ckpt_epoch_14.pth ...
accuracy on the test set: 0.885
```

图 21: 加入数据增广的预处理变换后模型的测试集准确率

加入数据增广预处理变换训练模型后，测试集的准确率为 88.5%，比验证集的最高准确率低 0.3%，验证集最高准确率与测试集准确率极为接近，说明泛化能力强，有效抑制了过拟合。

2.4 探究空间变换网络 (STN) 的作用

训练加入空间变换网络的模型：

执行 `python train.py -ckpt_path checkpoints/stn -bn -stn`

```
(demo) D:\college\media\HW2-B\code>python train.py --ckpt_path checkpoints/stn -bn -stn
training...
Epoch 1/15: 100% | 65/65 [00:07<00:00, 8.76it/s]
Epoch 01: loss = 2.018, accuracy on validation set = 0.585
Model saved in checkpoints/stn\ckpt_epoch_1.pth

Epoch 2/15: 100% | 65/65 [00:22<00:00, 2.84it/s]
Epoch 02: loss = 0.843, accuracy on validation set = 0.765
Model saved in checkpoints/stn\ckpt_epoch_2.pth

Epoch 3/15: 100% | 65/65 [00:27<00:00, 2.35it/s]
Epoch 03: loss = 0.484, accuracy on validation set = 0.783
Model saved in checkpoints/stn\ckpt_epoch_3.pth

Epoch 4/15: 100% | 65/65 [00:05<00:00, 12.36it/s]
Epoch 04: loss = 0.322, accuracy on validation set = 0.844
Model saved in checkpoints/stn\ckpt_epoch_4.pth

Epoch 5/15: 100% | 65/65 [00:06<00:00, 9.94it/s]
Epoch 05: loss = 0.231, accuracy on validation set = 0.852
Model saved in checkpoints/stn\ckpt_epoch_5.pth

Epoch 6/15: 100% | 65/65 [00:08<00:00, 8.02it/s]
Epoch 06: loss = 0.148, accuracy on validation set = 0.863
Model saved in checkpoints/stn\ckpt_epoch_6.pth

Epoch 7/15: 100% | 65/65 [00:07<00:00, 8.63it/s]
Epoch 07: loss = 0.090, accuracy on validation set = 0.900
Model saved in checkpoints/stn\ckpt_epoch_7.pth

Epoch 8/15: 100% | 65/65 [00:06<00:00, 9.39it/s]
Epoch 08: loss = 0.052, accuracy on validation set = 0.877
Model saved in checkpoints/stn\ckpt_epoch_8.pth

Epoch 9/15: 100% | 65/65 [00:08<00:00, 8.09it/s]
Epoch 09: loss = 0.041, accuracy on validation set = 0.894
Model saved in checkpoints/stn\ckpt_epoch_9.pth

Epoch 10/15: 100% | 65/65 [00:07<00:00, 9.01it/s]
Epoch 10: loss = 0.021, accuracy on validation set = 0.910
Model saved in checkpoints/stn\ckpt_epoch_10.pth

Epoch 11/15: 100% | 65/65 [00:05<00:00, 11.16it/s]
Epoch 11: loss = 0.029, accuracy on validation set = 0.904
Model saved in checkpoints/stn\ckpt_epoch_11.pth

Epoch 12/15: 100% | 65/65 [00:05<00:00, 12.02it/s]
Epoch 12: loss = 0.058, accuracy on validation set = 0.835
Model saved in checkpoints/stn\ckpt_epoch_12.pth

Epoch 13/15: 100% | 65/65 [00:07<00:00, 8.95it/s]
Epoch 13: loss = 0.161, accuracy on validation set = 0.815
Model saved in checkpoints/stn\ckpt_epoch_13.pth

Epoch 14/15: 100% | 65/65 [00:10<00:00, 6.11it/s]
Epoch 14: loss = 0.124, accuracy on validation set = 0.883
Model saved in checkpoints/stn\ckpt_epoch_14.pth

Epoch 15/15: 100% | 65/65 [00:07<00:00, 8.61it/s]
Epoch 15: loss = 0.068, accuracy on validation set = 0.921
Model saved in checkpoints/stn\ckpt_epoch_15.pth
```

图 22: 加入空间变换网络后模型的训练过程

加入空间变换网络 (STN) 训练模型时，验证集的准确率最高为 92.1%，较默认配置时准确率有所提升，但损失函数曲线的波动较大。由此可知，空间变换网络有效增强了模型的空间不变性，模型性能得到提升。但由于数据集较小，过拟合问题仍然存在。

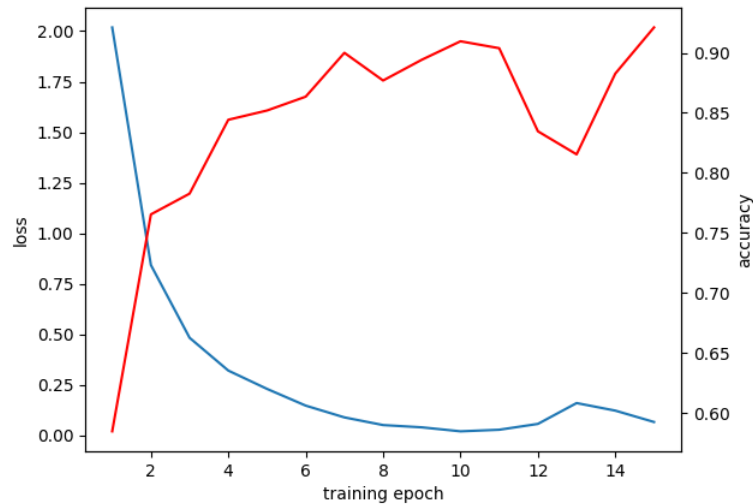


图 23: 加入空间变换网络后模型的损失函数

最优模型为验证集错误率最低点，由损失函数曲线可知加入空间变换网络后轮数 epoch=15 为最优模型。

测试加入空间变换网络的模型：

执行 `python test.py -ckpt_path checkpoints/stn -epoch 15`

```
(demo) D:\college\media\HW2-B\code>python test.py --ckpt_path checkpoints/stn --epoch 15
[Info] loading checkpoint from checkpoints/stn\ckpt_epoch_15.pth ...
accuracy on the test set: 0.909
```

图 24: 加入数据增广的预处理变换后模型的测试集准确率

加入空间变换网络 (STN) 训练模型后，测试集的准确率为 90.9%，比验证集的最高准确率低 1.2%，较默认配置情况有所提升，进一步证明模型性能得到了提示。

2.5 可视化

通过 `-ckpt_path` 和 `-epoch` 指定选取启用 batch normalization 的模型，轮数使用默认值 `epoch=15`。

2.5.1 卷积核

用如下命令语句可视化第 0 层卷积层的卷积核：

```
python visualize.py -type filter -ckpt_path checkpoints/bn -layer_idx 0
```

```
(demo) D:\college\media\HW2-B\code>python visualize.py --type filter --ckpt_path checkpoints/bn --layer_idx 0
[Info] loading checkpoint from checkpoints/bn/ckpt_epoch_15.pth ...
[ WARN:0@7.633] global loadsave.cpp:848 cv::imwrite_ Unsupported depth image for selected encoder is fallbacked to CV_8U.
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
Results are saved as visualized/filter/filter_layer_0.jpg
```

图 25: 执行卷积核可视化的命令

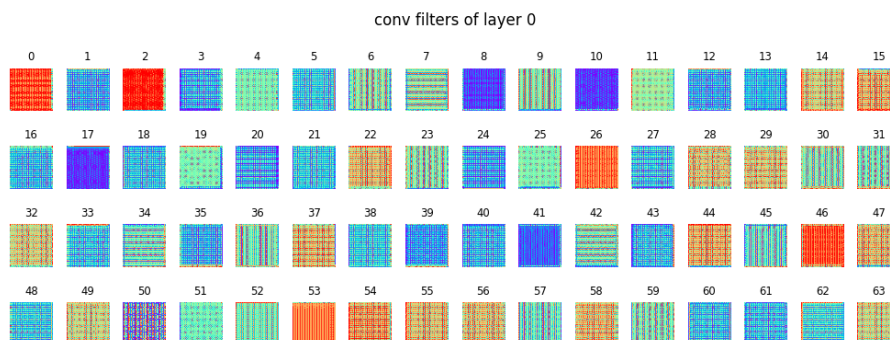


图 26: 卷积核可视化结果

由图可知，浅层卷积核的最大响应呈现简单的颜色和方向。

2.5.2 特征图

用如下命令语句可视化第 50 张图像第 1 层卷积层的输出特征图：

```
python visualize.py --type feature --ckpt_path checkpoints/bn --layer_idx 1  
--image_idx 50
```

```
(demo) D:\college\media\WM2-B\code>python visualize.py --type feature --ckpt_path checkpoints/bn --layer_idx 1 --image_idx 50  
[Info] loading checkpoint from checkpoints/bn/ckpt_epoch_15.pth ...  
[ WARN:0@5.795] global loadsave.cpp:848 cv::imwrite_ Unsupported depth image for selected encoder is fallbacked to CV_8U.  
libpng warning: iCCP: known incorrect sRGB profile  
libpng warning: iCCP: known incorrect sRGB profile  
libpng warning: iCCP: known incorrect sRGB profile  
libpng warning: iCCP: known incorrect sRGB profile  
libpng warning: iCCP: known incorrect sRGB profile  
Results are saved as visualized/feature/layer_1/feature_map.jpg
```

图 27: 执行特征图可视化的命令

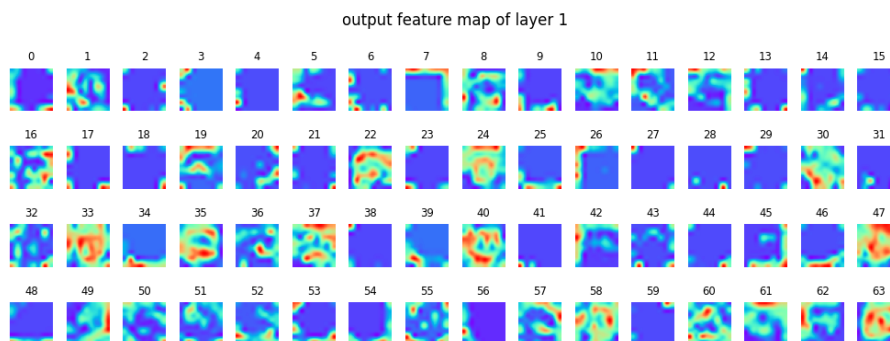


图 28: 特征图可视化结果

layer_idx 取值范围为 [0, 1, 2, 3, 4], 不同卷积核提取的特征不同：层数越深，特征就越抽象。由图可知层数为 1 时特征较为模糊抽象，但仍能辨认出部分轮廓。

2.5.3 t-SNE 降维后的最后一层隐藏层的输出特征

用如下命令语句可视化 t-SNE 最后一层隐藏层的输出特征：

```
python visualize.py -type tsne -ckpt_path checkpoints/bn
```

```
(demo) D:\college\media\HW2-B\code>python visualize.py --type tsne --ckpt_path checkpoints/bn
[Info] loading checkpoint from checkpoints/bn\ckpt_epoch_15.pth ...
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
Results are saved as visualized/tsne\tsne.jpg
```

图 29: 执行 t-SNE 最后一层隐藏层的输出特征可视化的命令

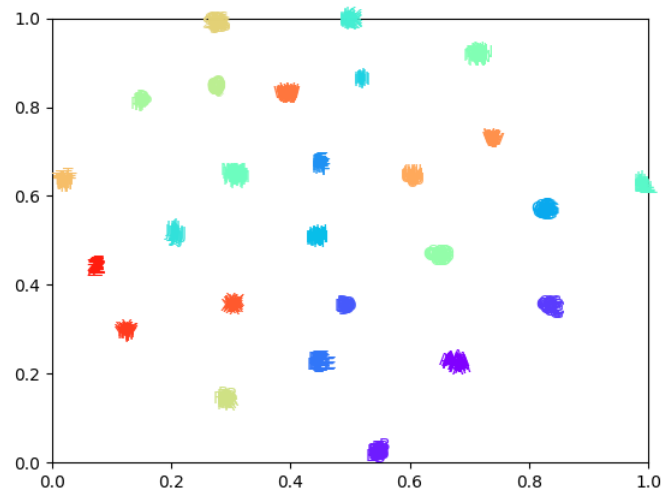


图 30: t-SNE 最后一层隐藏层的输出特征可视化结果

最后一层隐藏层的输出特征降维到 2 维后绘制的分类情况图表明，26 类交通标志图像的每一类样本点都在图中呈集聚状态，证明该卷积神经网络分类任务的实现情况较好。

2.5.4 空间变换网络学习到的空间变换

用如下命令语句可视化空间变换网络 (STN):

```
python visualize.py -type stn -ckpt_path checkpoints/stn
```

```
(demo) D:\college\media\HW2-B\code>python visualize.py --type stn --ckpt_path checkpoints/stn
[Info] loading checkpoint from checkpoints/stn\ckpt_epoch_15.pth ...
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
libpng warning: iCCP: known incorrect sRGB profile
Results are saved as visualized/stn\stn.jpg
```

图 31: 执行空间变换网络 (STN) 可视化的命令

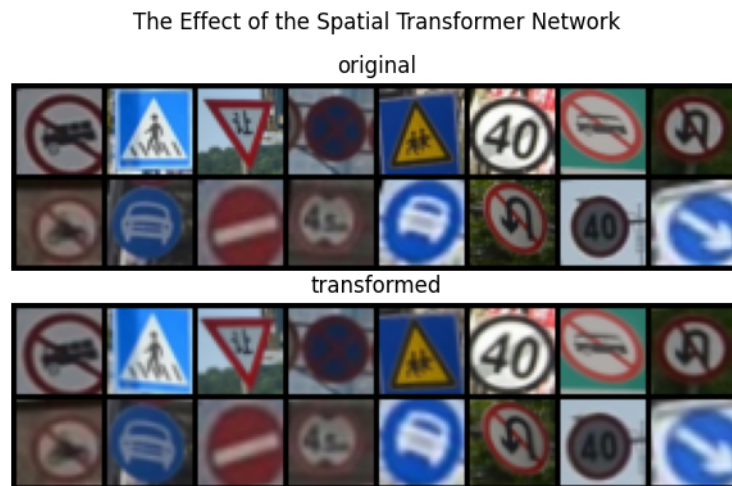


图 32: 空间变换网络 (STN) 可视化结果

输出了一组原始图像和经过空间变换网络 (STN) 处理后图像的对比, 可以看出空间变化网络 (STN) 成功在不改变原有图像核心特征的情况下, 对图像的位置、大小、方向等进行了微调, 以增强模型的空间不变性。